cm93depth_pi

CM93 and S-57 chart depth, look-ahead and grounding alarm plugin for OpenCPN

Technical and User Manual

Plugin version 1.5.1 | OpenCPN 5.8 and later (plugin API 1.18)

Windows (32/64-bit), Linux (x86_64), Raspberry Pi (arm64, armhf)

Charted depths are values below chart datum from CM93 and S-57 ENC data. The plugin is an aid to navigation and not a substitute for a sounder, a lookout or official charts.

Contents

1 Purpose and scope **3**
1.1 What the depths mean 3
2 System overview **4**
3 Installation **5**
3.1 Compatibility 5
3.2 Building the plugin package 5
3.3 Installing in OpenCPN 5
3.4 First set-up 5
4 How the charts are read **6**
4.1 Chart bands: CM93 and S-57 ENC 6
4.2 Chart objects used 7
4.3 Dangers with no charted depth, gaps and smaller-scale dangers 7
4.4 Soundings as OpenCPN draws them 7
4.5 Land 7
4.6 S-57 ENC cells 8
5 Depth at the vessel - search modes **9**
5.1 Interpolation in detail 9
6 Look-ahead **10**
6.1 The zone 10
6.2 Course and length 10
6.3 Which chart along the zone 10
6.4 What is found 10
6.5 Display 11
6.6 Choosing the safety depth 11
6.7 Limitations 11
7 Grounding alarm **12**
7.1 When it sounds 12
7.2 States and operator actions 12
7.3 Switching the alarm off 12
7.4 Sound 12
8 Outputs to other devices **13**
8.1 Routes, rate and validity 13
8.2 $PCMLD - depth at the vessel 13
8.3 $PCMLA - look-ahead 13
8.4 $IIALR - alarm (optional) 14
8.5 Integration notes 14
9 Display window **15**
9.1 Copy report 15
10 Settings reference **16**
10.1 Charts and search 16
10.2 Look-ahead 16
10.3 Output and chart 17
10.4 Alarm 17
10.5 Configuration-file-only keys 17
11 Recommended settings **18**
12 Troubleshooting **19**
13 Limitations **20**

1 Purpose and scope

cm93depth_pi reads the vector charts used by OpenCPN - CM93 cells and unencrypted S-57 ENC cells - at the vessel's position and ahead of it, and turns them into numbers other equipment can use. Where both kinds cover the same water, the most detailed chart is used. It does four things:

- **Depth at the vessel**: the charted depth where the vessel is, in one of four search modes (depth area here, whole cell, radius, interpolated from the nearest soundings).
- **Look-ahead**: the least charted depth in a zone ahead along the course, and the **first shoal** - the nearest point ahead shallower than a safety depth, land, or a danger with no charted depth - with its distance, time and bearing.
- **Grounding alarm** on the OpenCPN PC: sound and an always-on-top alarm window when a shoal is close ahead or the depth at the vessel is below the safety depth, with Silence, Acknowledge and a watch-officer off switch.
- **NMEA 0183 output** to other devices: `$PCMLD` (depth at the vessel), `$PCMLA` (look-ahead) and optionally `$IIALR` (alarm), on a serial port or through an OpenCPN output connection.

It also shows the results in a small window, draws the look-ahead zone on the chart and can mark the least-depth position.

1.1 What the depths mean

- All depths are **charted values below chart datum** (approximately the lowest astronomical tide). No tide or height of tide is applied.
- CM93 data is old and carries no survey-quality information. ENC cells are current official data, but the plugin does not use their survey-quality (CATZOC) information either. Interpolated depths are estimates between soundings, not measurements.
- Where the chart has no depth for something (a reef, an obstruction, a sand flat) the plugin takes the cautious value described in section 4, normally 0 m.
- The plugin knows only what is charted. A feature missing from the charts is invisible to it.

Use the plugin as an aid alongside the echo sounder, a proper lookout and the best available charts. It is not certified navigation equipment.

2 System overview

Inputs, processing and outputs:

	What	Detail
Input	Position fix	From OpenCPN: latitude, longitude, COG, SOG and true heading (if any). A fix not updated for the 'Fix lost after' time is treated as lost (status V).
Input	CM93 charts	The CM93 v2 folder used by OpenCPN (found automatically or set in Preferences). Cells are read directly; compressed .xz cells are supported.
Input	S-57 ENC cells	Unencrypted *.000 base cells with their *.001, *.002 ... updates, found in OpenCPN's chart directories (and any extra folders set). Read directly, not through OpenCPN's SENC files. Encrypted S-63 cells cannot be read.
Processing	Every update interval	(1 s default) Depth at the vessel; look-ahead; alarm evaluation; output; window and chart refresh.
Output	Display window	Depth figure, caption, shoal warning; full details on request.
Output	Chart overlay	Dashed look-ahead zone (green clear, red with a shoal), ring on the first shoal; optional temporary mark at the least-depth position.
Output	NMEA 0183	`$PCMLD`, `$PCMLA`, optional `$IIALR`; serial port (4800-115200 baud) or an OpenCPN output connection (serial or network/UDP).
Output	Alarm	Sound (OpenCPN sound system) and alarm window on the OpenCPN PC.

3 Installation

3.1 Compatibility

- OpenCPN **5.8 or later** (plugin API 1.18); uses nothing newer.
- A plugin build must match the OpenCPN it runs in: 32- or 64-bit, and the wxWidgets version (3.2 for current OpenCPN). One build serves every PC with the same OpenCPN type.
- Raspberry Pi OS / Debian 12 'bookworm' ship OpenCPN 5.6 (too old); use the Debian 13 'trixie' based release (OpenCPN 5.10). The Flatpak OpenCPN is not supported yet.

3.2 Building the plugin package

The same source package serves all systems. Building produces a **plugin tarball** (`.tar.gz`) that OpenCPN installs itself.

System	Needs	Command	Result
Windows	Visual Studio 2022 (Community) with 'Desktop development with C++'; 7-Zip; the target OpenCPN installed	double-click `windows\build_windows.bat`	`windows\dist\cm93depth_pi-<ver>-0_msvc-wx32-10-x86.tar.gz` (32-bit) or `..._msvc-64-10-x86_64` (64-bit)
Linux Mint / Ubuntu / Debian / Raspberry Pi OS	OpenCPN 5.8+ from apt or the OpenCPN PPA; the script installs build tools with apt	`bash ~/cm93depth_pi/linux/build_linux.sh`	`linux/dist/cm93depth_pi-<ver>-0_<distro>-<release>-<arch>.tar.gz`

The Windows script finds OpenCPN (installer registry entry, Program Files, or asks), detects 32/64-bit and the wxWidgets version, downloads the matching wxWidgets development files once (cached in `%LOCALAPPDATA%\cm93depth_pi-build`), builds an import library from opencpn.exe and compiles. The Linux script installs cmake, g++, the wxWidgets 3.2 GTK development files, liblzma, OpenGL headers and lsb-release, then compiles. Both write the tarball and change nothing in OpenCPN unless run with `-Install` / `--install` (development use).

3.3 Installing in OpenCPN

- Options > Plugins > **Import plugin...**, choose the tarball. OpenCPN reports 'successfully imported'.
- Enable **CM93 Least Depth** in the plugin list. A toolbar button (sloping seabed with a red arrow) toggles the display window.
- Updating: import the newer tarball; it replaces the old one. Uninstalling: select the plugin in Options > Plugins and choose Uninstall.
- Files go to `%LOCALAPPDATA%\opencpn\plugins\` (Windows) or `~/.local/lib/opencpn/` (Linux).

3.4 First set-up

- Settings... (in the window) or Options > Plugins > Preferences.
- Charts and search tab: choose the chart sources (CM93, S-57 ENC, or both; section 4.6). For CM93 leave 'Find CM93 automatically' ticked, or point to the folder holding the 8-digit cell folders and CM93OBJ.DIC. ENC cells are found in OpenCPN's chart directories by themselves. Choose the search mode (section 5).
- Look-ahead tab: set the **safety depth** for your vessel (section 6.6).
- Output and chart tab: choose where the sentences go (section 8).
- Linux: to use a serial port your user must be in the `dialout` group (`sudo usermod -aG dialout $USER`, then log out and in).

4 How the charts are read

4.1 Chart bands: CM93 and S-57 ENC

CM93 divides the world into cells at eight scale bands. S-57 ENC cells each belong to one of six usage bands. The plugin puts both into one order of detail and always prefers the most detailed band that has data at the place in question, whichever kind of chart it is.

Band	Nominal scale	Cell size	Typical use
Z	1:20,000,000	40°	World
A	1:3,000,000	20°	Ocean
B	1:1,000,000	10°	Ocean / coastal approach
C	1:200,000	4°	Coastal
D	1:100,000	1°	Coastal
E	1:50,000	20'	Approach
F	1:20,000	20'	Harbour approach
G	1:7,500	20'	Harbour

ENC band	Usage	Typical compilation scale
1	Overview	smaller than 1:1,500,000
2	General	1:350,000 - 1:1,500,000
3	Coastal	1:90,000 - 1:350,000
4	Approach	1:22,000 - 1:90,000
5	Harbour	1:4,000 - 1:22,000
6	Berthing	larger than 1:4,000

The usage band is the cell's intended usage (DSID INTU), or, if that is missing, follows from its compilation scale. Combined order, least detailed first:

Order	Band	Chart
1 (least detailed)	Z	CM93 1:20,000,000
2	A	CM93 1:3,000,000
3	1	ENC Overview
4	B	CM93 1:1,000,000
5	2	ENC General
6	C	CM93 1:200,000
7	3	ENC Coastal
8	D	CM93 1:100,000
9	E	CM93 1:50,000
10	4	ENC Approach
11	F	CM93 1:20,000
12	5	ENC Harbour
13	G	CM93 1:7,500
14 (most detailed)	6	ENC Berthing

So a harbour ENC (band 5) is used in preference to CM93 A to F but not to CM93 G; a berthing ENC (6) beats everything; where a detailed CM93 cell's coverage stops, a coastal or approach ENC fills in before the smaller CM93 scales. Where two ENC cells of the same band overlap, the one with the larger compilation scale is tried first, and in depth-area mode the shallower depth area of the two counts.

The band used for each result is reported - CM93 by its letter, ENC by its digit - in field 14 of `$PCMLD`, field 15 of `$PCMLA` and the window (with the ENC cell's name). 'Scale band' in Preferences can force one CM93 band (ENC cells are then not used); Auto is recommended.

4.2 Chart objects used

Object (S-57 class)	How it is used
Depth area DEPARE (incl. CM93 intertidal ITDARE), dredged area DRGARE	Depth band DRVAL1..DRVAL2. With no DRVAL1 (charted but no depth, e.g. a sand or mud flat): 0 to 0 m. Intertidal area with only DRVAL2: that value.
Sounding SOUNDG	Spot depth. Read exactly as OpenCPN draws them (see 4.4).
Depth contour DEPCNT	VALDCO as a depth along the line (cell / radius modes, look-ahead).
Rock UWTROC, wreck WRECKS, obstruction OBSTRN	Danger. VALSOU if charted. No VALSOU but WATLEV 'dries', 'covers and uncovers', 'awash' or 'partly submerged': 0 m. No VALSOU otherwise (e.g. a submerged reef): **no charted depth** - see 4.3. Obstructions with CATOBS foul area, foul ground or fish haven are not dangers.
Seabed area SBDARE (e.g. sand)	Dries / covers / awash (WATLEV): 0 m. No depth and no water level, where no depth area is charted: 0 m. Inside a depth area: ignored (nature of the bottom only).
Land area LNDARE	Land: no depth; the vessel on land gives 'LAND' and status V; land in the look-ahead zone is a shoal; land between the vessel and a sounding hides that sounding.
Coverage M_COVR	Which part of a cell holds data (CATCOV 1).

4.3 Dangers with no charted depth, gaps and smaller-scale dangers

- **Undepthed danger under the vessel** (depth area here and interpolated modes): taken as 0 m (default) or, by setting, as the least depth of the surrounding depth area (the ECDIS rule, IHO S-52 DEPVAL02).
- **Gaps**: depth areas (CM93 and ENC) cover all charted water. A position inside a hole in the depth areas, closed in on all sides at the most detailed band that has depth areas round it, is on something charted without a depth (reef, rock, islet, unsurveyed area; in an ENC also a pontoon, floating dock or hulk, which ENCs leave out of the depth areas). It is treated as an undepthed danger; source letter **U**. In an ENC only positions inside the cell's coverage can be gaps.
- **Smaller-scale dangers**: rocks, wrecks and obstructions charted only on a smaller-scale chart than the one the depth comes from still count (default), CM93 or ENC. CM93's small-scale cells carry obstruction areas its detailed cells sometimes lack. The shallowest danger wins.
- **Danger areas overruled by detailed soundings**: small-scale charts also mark whole zones as obstruction areas (a B-scale CM93 cell's notes say, for example, that 'numerous structures and submerged obstructions exist in certain areas indicated on this chart; see larger scale charts'). A rock, wreck or obstruction **area** is therefore not counted where a **more detailed** chart has a sounding inside it, within 0.25 nm (setting) of the vessel - or of where the look-ahead zone meets it - that is at least the safety depth deep (and deeper than the danger's own depth). A real reef that the detailed chart has missed has no deep detailed soundings inside its outline beside the vessel, so it still counts. Point dangers always count. The window and Copy report say when an area was overruled and by which sounding. Example: an undepthed obstruction area on the 1:1,000,000 chart over Burgess Bank, where the 1:100,000 and 1:50,000 charts sound 10.8 m and 7.4 m within 0.14 nm in a 10-15 m area: the depth is 10 m, not 0 m.

4.4 Soundings as OpenCPN draws them

OpenCPN's CM93 loader keeps only the first sounding of a record holding several (the rest are overwritten by the next record). The plugin does the same by default, so it uses exactly the soundings on the chart. Config key `OpenCpnSoundings=0` uses every stored sounding instead.

4.5 Land

At the most detailed chart band that has either water or land at the vessel's position, land there (and no water) means the vessel is on charted land: no depth, window 'LAND', status V, and (if enabled) the grounding alarm.

4.6 S-57 ENC cells

- **Which cells**: every `*.000` file in OpenCPN's chart directories (Options > Charts > Chart Files) and their sub-folders, plus any 'More ENC folders' set in Preferences. The list is checked once a minute and the cells indexed again when OpenCPN's chart directories change.
- **Index**: for each cell the plugin records its name, usage band, compilation scale and extent. The first time, it reads each cell's header in small steps on its timer (setting 'ENCIndexMs', 150 ms per tick), so OpenCPN stays responsive; cells already indexed are used meanwhile. Where an exchange set has a `CATALOG.031`, cells listed there with their extent are indexed without being read. The index is kept in `plugins/cm93depth_pi/enc_index.txt` in OpenCPN's data folder; later starts re-read only new or changed cells. **Rescan ENC cells** in Preferences does it at once.
- **Updates**: `*.001`, `*.002` ... beside the base cell are applied in order (records inserted, deleted and modified; attributes, pointers and coordinates), as OpenCPN does. Updates stop at the first one missing or out of sequence, or for another edition. A cell cancelled by an update (edition 0) is still used, as OpenCPN still shows it. 'Copy report' lists the updates applied and any problem.
- **Units**: positions from the cell's coordinate factor; depths and soundings in metres (cells in feet or fathoms are converted).
- **Encrypted S-63 cells** (oeSENC, o-charts and other encrypted services) cannot be read and are skipped; Preferences shows how many cells could not be read.
- **Same objects, same rules**: the object classes and rules of 4.2 and 4.3 apply unchanged; ENC soundings are used exactly as stored.

5 Depth at the vessel - search modes

Mode (Preferences)	Letter	Result	Use
Depth area at the vessel (default)	H	Minimum (DRVAL1) of the depth area containing the vessel, at the most detailed scale that has one there. Dangers inside the area that are undepthed or shallower are counted; a danger under the vessel limits the depth.	Robust everywhere; conservative (the shallow end of the band).
Interpolated from the nearest soundings	I	Depth estimated from the four nearest soundings, limited to the depth band at the vessel (5.1).	Best estimate of the actual charted depth; recommended for the Sea State Monitor.
Within a radius of the vessel	R	Least depth of any object within the radius.	Warning of anything shallow nearby.
Whole cell under the vessel	C	Least depth anywhere in the cell.	Rarely useful near a coast (usually 0 m).

5.1 Interpolation in detail

- **Pool**: soundings within the radius from every scale band (the cell under the vessel and its eight neighbours at each band). The same sounding repeated at several bands (within ~20 m and 0.05 m) is counted once.
- **Land check** (default on): soundings with charted land between them and the vessel (the far side of a headland, behind an island) are left out, each tested against the coastline of its own scale band. A coastline that covers the vessel itself is not used for the test.
- **Selection**: the four nearest. Option: one from each quadrant (NE, SE, SW, NW) first.
- **Estimate**: inverse-distance weighting, power 2. A sounding within ~2 m is taken outright.
- **Limits**: the estimate is held within the DRVAL1..DRVAL2 band of the depth area at the vessel (an area with no depth is 0 to 0 m), then at or below the depth of any danger under the vessel. The window shows the raw estimate when it was limited.
- `$PCMLD` field 6 gives the distance to the nearest sounding used, field 13 the farthest.

Radius: in well-charted water 2-10 nm is enough; in sparsely charted water up to 50 nm works with the land check on. A radius larger than needed does no harm because only the nearest four are used.

6 Look-ahead

6.1 The zone

Each update a zone is built ahead of the vessel along the course:

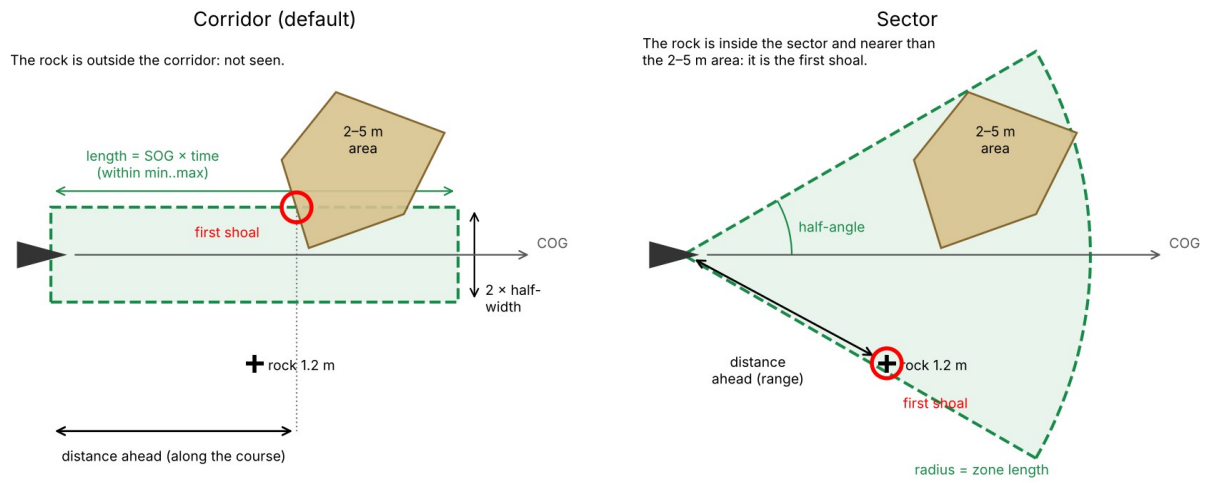


Figure 1. Look-ahead zones. The first shoal is the nearest point ahead, inside the zone, that is shallower than the safety depth, land, or a danger with no charted depth.

Item	Corridor (default)	Sector
Shape	Strip along the course, 2 x half-width wide	Pie slice of ± half-angle about the course (max 90°)
Length	Zone length	Radius = zone length
'Distance ahead'	Along the course	Range from the vessel
Best for	Holding a course; passage	Pilotage, manoeuvring, poorly charted waters

6.2 Course and length

- **Course**: COG while SOG is at least the minimum SOG (0.5 kn). Below that the true heading is used (if OpenCPN has one), the zone has its shortest length and no times are given. With neither: status V, 'No course'.
- **Length, time basis** (default): SOG x time ahead (10 min), never shorter than the shortest zone (0.5 nm) or longer than the longest zone (5 nm). Examples: 2 kn 0.5 nm; 6 kn 1.0 nm; 15 kn 2.5 nm; 30 kn 5 nm.
- **Length, distance basis**: a fixed length. Hard limit 50 nm either way.

6.3 Which chart along the zone

The zone is cut into slices of about 0.1 nm (at most 40). For each slice the most detailed chart band (CM93 or ENC) that charts all its test points (water, land or a gap) is used; where none does, every band charting part of it. Neighbouring slices with the same choice are searched together. This follows the most detailed chart along the way, as a chart display does, without mixing in generalised small-scale shoals where a detailed chart exists. Rocks, wrecks and obstructions from smaller-scale charts are added (4.3).

6.4 What is found

- Objects are clipped to the zone **exactly**, edge by edge, so a shoal that only touches the edge of the corridor is found at the point where it enters.

- **Gaps** in the depth areas are found exactly where the zone first enters them, however small: the zone can only enter a gap across a depth-area or land edge, so each such edge crossing the zone is tested on both sides.
- **First shoal**: the nearest point ahead that is charted shallower than the safety depth (any object), land, or - with the option on (default) - a danger with no charted depth or a gap. At equal distance land comes first, then the shallower.
- **Least depth in the zone**: the shallowest charted value of any object in the zone (land not counted).
- **Undepthed dangers**: their number and the nearest.

6.5 Display

The zone is drawn dashed on the chart, green when clear and red with a ring at the first shoal (works with and without OpenGL; can be switched off). The window shows 'Shoal ahead (< 5.0 m): 0.85 nm, 8.5 min, 088° T' or 'Clear ahead', the least depth ahead, the zone used and the scales along it.

6.6 Choosing the safety depth

Safety depth = draft + under-keel clearance + allowance for chart error and sea state. For a yacht drawing 2.0 m: 2.0 + 1.0 + 2.0 = 5 m is a reasonable start. Remember depths are below chart datum: at most states of tide there is more water, but never assume it.

6.7 Limitations

- Only what is charted. A corridor narrower than the cross-track error, leeway or a planned turn misses what lies outside it; use a wider corridor or a sector when manoeuvring.
- A point danger is found only if it lies inside the zone.
- The zone follows COG, which lags the heading in a turn.

7 Grounding alarm

7.1 When it sounds

- The look-ahead's first shoal is within **6 minutes** at the current SOG, or within **0.2 nm** at any speed (distance only when the course comes from the heading).
- The depth **at the vessel** is below the safety depth (depth area here and interpolated modes only), or the vessel is **on charted land** (any mode).
- Only with a valid position fix. The alarm time should be shorter than the look-ahead time, since only the zone is searched.

7.2 States and operator actions

State	Sound	Alarm window	Leaves the state when
Active	On entry and every 5 s	Flashing red; 'Sounding'	Silenced, acknowledged, or the condition clears
Silenced	Muted for 30 s	Flashing red; 'Silenced for n s'	After 30 s: back to Active if not acknowledged
Acknowledged	None	Steady amber	Gets worse (time to the shoal halves, or depth here below the safety depth): Active again. Condition clears: closes.
Clear	None	Closed	A condition appears: Active at once

- **Silence**: mutes for the silence period (IMO bridge alert practice: 30 s).
- **Acknowledge**: accepts the alarm; no sound while the condition lasts.
- The window cannot be closed without acknowledging.

7.3 Switching the alarm off

For navigable water that is shallower than the safety depth (a sheltered anchorage, a known channel) the watch officer can switch the alarm off completely: the **Alarm: on** button in the display window, or **Switch alarm off...** in the alarm window. Both ask first (No is the default).

- While off: no sound or alarm window; depth, look-ahead, chart zone and sentences continue. The window shows **GROUNDING ALARM OFF** in amber and the button reads **Alarm: OFF**.
- On again: click **Alarm: OFF**; or automatically after a set time (Alarm tab, 0 = never); and always when OpenCPN starts (the off state is never saved).
- Switching off and on is recorded in the OpenCPN log.
- Alternative for a long stretch: lower the safety depth instead, keeping protection against anything shallower still.

7.4 Sound

OpenCPN's own `sounds/beep_ssl.wav` through OpenCPN's sound system, or a WAV file chosen on the Alarm tab (Test button). If no sound can be played the system bell is used.

8 Outputs to other devices

8.1 Routes, rate and validity

- **Send via**: Off (default) / the plugin's own **serial port** (port and baud) / an **OpenCPN output connection** (serial or network, e.g. UDP to the autopilot or Sea State Monitor). Use the OpenCPN connection when OpenCPN already has the port open. The connection must have output enabled, and any output filter must pass PCMLD, PCMLA (and ALR if used).
- **Rate**: one `$PCMLD` and one `$PCMLA` per update interval (1 s). At 4800 baud that is about 30 % of the link; use 38400 baud or a network connection if the link is shared.
- **Invalid data**: with 'Keep sending with status V' (default) a sentence goes out every interval even without valid data, status V and empty value fields, so a receiver can tell 'no data' from 'link dead'. Receivers should treat status V, or no sentence for a few intervals, as no chart data.
- All sentences carry the standard NMEA checksum and stay within the 82-character limit.

8.2 $PCMLD - depth at the vessel

```
$PCMLD,15.0,4307.000,S,14716.700,E,0.00,,15.0,20.0,A,0,H,,G,A*hh (depth area here)
$PCMLD,46.2,4336.790,S,14716.138,E,0.29,,50.0,100,I,0,I,0.45,E,A*hh (interpolated)
```

#	Field	Example
1	Depth, m below chart datum (negative = drying). H: minimum of the depth area at the vessel; I: interpolated depth at the vessel; R/C: least depth in scope	15.0
2-3	Latitude of the depth, ddmm.mmm, N/S (H, I: the vessel)	4307.000,S
4-5	Longitude, dddmm.mmm, E/W	14716.700,E
6	Range from the vessel, nm (H: 0.00; I: nearest sounding used)	0.29
7	True bearing, ° (empty at range 0 and in mode I)	090
8	DRVAL1 of the depth area at the vessel, m	15.0
9	DRVAL2 of that area, m (empty if none)	20.0
10	Source: S sounding, A depth area / seabed area, D dredged area, O obstruction, W wreck, R rock, C contour, I interpolated, U gap in the depth areas	A
11	Dangers with no charted depth in scope (H, I: also those charted shallower)	0
12	Mode: H depth area here, I interpolated, R radius, C whole cell	H
13	Radius, nm (R); farthest sounding used (I); empty otherwise	0.45
14	Chart band used: CM93 Z..G, or ENC usage band 1..6	G
15	Status: A valid; V invalid (no/stale fix, no chart, no depth, on land)	A

Resolution: depths 0.1 m (whole metres from 100 m); distances 0.01 nm below 10 nm. The address PCMLD can be changed (P + four capitals/digits).

8.3 $PCMLA - look-ahead

```
$PCMLA,1.0,1.92,090,S,5.0,1.00,10.0,079,W,1,1.00,C,2.00,0.25,E,A*hh
```

#	Field	Example
1	Least charted depth in the zone, m (land not counted)	1.0
2	Range to it, nm	1.92
3	True bearing to it, °	090
4	Its source (as $PCMLD field 10)	S
5	Safety depth, m (the setting)	5.0
6	Distance ahead to the first shoal, nm (along the course for a corridor, range for a sector); empty = none in the zone	1.00
7	Time to reach it at the current SOG, min (empty below the minimum SOG)	10.0
8	True bearing to it, °	079

#	Field	Example
9	Its source: S A D O W R C, L land, U gap	W
10	Dangers with no charted depth in the zone (incl. gaps)	1
11	Distance ahead to the nearest of them, nm	1.00
12	Zone: C corridor, S sector	C
13	Zone length (sector radius), nm	2.00
14	Corridor half-width, nm, or sector half-angle, °	0.25
15	Least detailed chart band used along the zone (as $PCMLD field 14)	E
16	Status: A valid; V invalid (no/stale fix, no chart, no course)	A

For a control system the key fields are 6 and 7 (how far, how soon), 8 (which side) and 16. Worst-case length 72 characters.

8.4 $IIALR - alarm (optional)

```
$IIALR,135407.25,001,A,V,SHOAL AHEAD: 1.00 nm 5.0 min 088 T*hh
```

IEC 61162-1 alarm sentence: UTC time, alarm number 001, condition A (threshold exceeded) / V, acknowledged A / V, text. Sent each interval while the alarm is raised and once (V) when it clears. Off by default (Alarm tab).

8.5 Integration notes

- **Heading Control Unit / HMI**: parse $PCMLA fields 6-8 and 16 for a time-to-shoal term; treat V or a missing sentence as 'no chart data', never as 'clear'.
- **Sea State Monitor (SSM)**: accepts `$PCMLD` from the HCU or directly from the chart PC over UDP. In mode I it uses field 1 when the nearest sounding (field 6) is within its limit; in modes H, R, C the middle of the depth band (fields 8-9). Set an OpenCPN UDP output connection to the SSM's address and port, select it as the plugin's output, and set the chart PC's address in the SSM firmware.

9 Display window

- **Minimised** (default): the depth (or LAND), what it is ('Interpolated chart depth', 'Over an obstruction, no charted depth'...), a red shoal warning when the look-ahead has one, an amber 'GROUNDING ALARM OFF' when switched off, and the **Enlarge** and **Alarm** buttons.
- **Enlarged**: soundings used with their scale bands and weights, the depth band, dangers, scales searched, look-ahead details, status, output status, the last sentences, **Copy report** and **Settings...**. The size chosen is remembered.
- Day / dusk / night colours follow OpenCPN's colour scheme.

9.1 Copy report

Copies a troubleshooting report for the current position to the clipboard. It starts with the CM93 folder and the ENC index (cells, unreadable ones). Then for each chart band: the CM93 cell files or the ENC cells whose extent holds the position (name, band, compilation scale, file, updates applied), read errors, coverage, the nearest six soundings, every depth area whose box holds the position (with DRVAL1/DRVAL2 and the band actually used) and **every other object** containing the position or within 0.1 nm, with its attributes; a 'gap' line if applicable; then the result itself - depth and source, band, soundings used, any limit, any danger under the vessel and the chart it comes from - and the look-ahead status.

10 Settings reference

All settings are in Preferences (four tabs) and are stored in OpenCPN's configuration file (`C:\ProgramData\opencpn\opencpn.ini` on Windows, `~/.opencpn/opencpn.conf` on Linux) under `[PlugIns/cm93depth_pi]`. Factory defaults are documented in the USER SETTINGS block at the top of `src/cm93depth_pi.cpp`.

10.1 Charts and search

Setting	Default	Range	Purpose / effect
Chart sources: CM93 charts	on		Read CM93 cells.
Chart sources: S-57 ENC cells	on		Read unencrypted ENC cells (4.6). With both on, the most detailed chart wins (4.1).
More ENC folders	empty	folders, ';' between	Besides OpenCPN's chart directories.
Rescan ENC cells (button)			Look for new or updated ENC cells now.
Find CM93 automatically / CM93 folder	automatic		Where the CM93 data is.
Scale band	Auto	Auto, Z..G	Force one CM93 band (ENC cells then unused; not recommended).
Offset east / north	0 m	±5000 m	Same as OpenCPN's CM93 offset; only if charts are shifted.
Search area (mode)	Depth area at the vessel	4 modes	Section 5.
Radius	1.0 nm	0.05-50	Mode R: search radius. Mode I: farthest sounding.
Interpolation: one sounding from each direction	off		Surround the vessel; can reach for distant soundings.
Interpolation: leave out soundings with land between	on		Stops a large radius reaching round a headland.
Over a rock / wreck / reef with no charted depth	0 m	0 m / ECDIS rule	Section 4.3.
Count dangers charted only at smaller scales	on		Section 4.3; off gives fewer alarms from generalised areas.
Danger areas overruled by a more detailed chart's soundings within	0.25 nm	0-2 (0 = never)	Section 4.3. Raise: fewer false 0 m readings from generalised zones; lower: needs a detailed sounding closer by. The sounding must be at least the safety depth (Look-ahead tab).
Count these chart objects	all on		Soundings, depth/dredged areas, dangers, contours (modes C, R).

10.2 Look-ahead

Setting	Default	Range	Purpose / effect
Look ahead ... send the look-ahead sentence	on		Look-ahead, zone drawing, $PCMLA, shoal alarm.
Zone	Corridor	Corridor / Sector	Section 6.1.
Length	Time	Time / Fixed distance	Section 6.2.
Time ahead	10 min	1-120	Raise: earlier warning, more water you may not reach.
Shortest zone	0.5 nm	0.05-50	Low speed and minimum on a time basis.
Fixed distance	2.0 nm	0.05-50	Distance basis.
Longest zone	5 nm	0.1-50	Upper limit either basis.

Setting	Default	Range	Purpose / effect
Corridor half-width	0.1 nm	0.01-5	Raise for cross-track error and leeway.
Sector half-angle	20°	1-90	Raise to cover more of the water you could steer into.
Safety depth	5.0 m	-5 to 100	Shallower = shoal. Draft + clearance + allowance.
Minimum SOG	0.5 kn	0.1-5	Below: heading used, no times.
Sentence address	PCMLA		Must differ from the depth sentence address.
Undepthed dangers count as shoals	on		Includes gaps.
Draw the zone and the first shoal	on		Chart overlay.

10.3 Output and chart

Setting	Default	Range	Purpose / effect
Send via	Off	Off / Serial / OpenCPN connection	Section 8.1.
Serial port, Baud rate	-, 4800	4800-115200	Plugin's own port.
OpenCPN connection	-	output connections	Send through OpenCPN.
Sentence address	PCMLD		P + four capitals/digits.
Update interval	1.0 s	0.2-10	Calculation and transmit rate.
Fix lost after	10 s	2-120	Older fix = status V.
Keep sending with status V	on		Section 8.1.
Mark the least-depth position, icon	on, triangle		Modes C and R only.

10.4 Alarm

Setting	Default	Range	Purpose / effect
Grounding alarm on this PC	on		Section 7.
Shoal ahead within	6 min	0.5-60	Keep below the look-ahead time.
or within	0.2 nm	0.01-5	Any speed.
Repeat sound every	5 s	1-120	While unacknowledged.
Silence mutes for	30 s	5-300	IMO practice 30 s.
Switched off: back on after	0 min (never)	0-720	Automatic re-arm.
Sound file	OpenCPN beep	WAV	Test button.
Also when the depth here is below the safety depth, or on land	on		Modes H and I.
Also send it as $IIALR	off		Section 8.4.

10.5 Configuration-file-only keys

Key	Default	Purpose
OpenCpnSoundings	1	0 = use every stored sounding, not only those OpenCPN draws.
ENCIndexMs	150	Milliseconds per timer tick spent building the ENC index (20-2000).
UseCM93, UseENC, ENCExtraDirs	1, 1, -	Chart sources (set in Preferences).
PanelCompact, PanelVisible, PanelX/Y	1, 1, -	Window state (set from the window).

11 Recommended settings

Starting points for a sailing yacht of about 2 m draft. Adjust the safety depth for your vessel.

Setting	Coastal passage	Pilotage / narrow waters	Offshore	Poorly charted (reefs)
Search mode	Interpolated	Interpolated	Depth area here	Interpolated
Radius	10 nm	5 nm	-	20-50 nm
Zone	Corridor	Sector	Corridor	Sector
Half-width / half-angle	0.1-0.2 nm	±25°	0.3-0.5 nm	±30°
Time ahead / longest zone	10 min / 5 nm	5 min / 2 nm	20 min / 10 nm	10 min / 5 nm
Shortest zone	0.5 nm	0.3 nm	1 nm	0.5 nm
Safety depth	5 m	4-5 m	10 m	6-8 m
Alarm within	6 min / 0.2 nm	3 min / 0.1 nm	10 min / 0.5 nm	8 min / 0.3 nm
Undepthed danger depth	0 m	0 m	0 m	0 m
Dangers from smaller scales	on	on	on	on

- Update interval 1 s; output to the autopilot and Sea State Monitor through an OpenCPN UDP output connection.
- In sheltered water shallower than the safety depth: lower the safety depth for the stretch, or switch the alarm off and let it re-arm after a set time.
- In waters known to have uncharted reefs nothing replaces a lookout: the plugin cannot see what CM93 does not chart.

12 Troubleshooting

Symptom	Cause and remedy
Plugin not in the list after import	Built for the other type of OpenCPN (32/64-bit, other release/architecture on Linux): build against the OpenCPN it is for. See the OpenCPN log (lines with cm93depth).
'Incompatible import plugin'	Tarball for the other OpenCPN type.
Windows link errors when building	Fixed from 1.3.1 (API classes are no longer exported). Use the latest source.
OpenCPN crashed when disabling the plugin with its window open	Fixed in 1.3.4.
Look-ahead: 'No course'	SOG below the minimum and no heading in OpenCPN. Normal when stopped or when the boat is placed by hand.
'No depth area charted' / unexpected depth	Use Copy report at the position: it shows every object and which scale was used.
Serial 'Access is denied' / permission denied	Another program or an OpenCPN connection has the port: send via that OpenCPN connection instead. Linux: dialout group.
No alarm sound	Test on the Alarm tab; check OpenCPN's sound settings and the PC's audio output.
Many alarms over open water / 0 m over charted water	Smaller-scale obstruction areas. Copy report shows the area (with its extent) and its chart band. Detailed soundings inside it overrule it within the set distance (4.3); otherwise switch off 'Count dangers charted only at smaller scales', or use the ECDIS rule.
ENC cells not used	Preferences > Charts and search shows the ENC index: 'indexing n of m' (wait), 'no ENC cells' (the folder is not among OpenCPN's chart directories: add it there or under 'More ENC folders'), or 'n unreadable' (encrypted S-63 cells, or damaged files; Copy report names the first). A CM93 band more detailed than the ENC wins by design (4.1).
New ENC updates not seen	Press Rescan ENC cells (or restart OpenCPN).

13 Limitations

- CM93 is old, unmaintained data with no survey-quality information; features may be missing or displaced. The plugin cannot detect what is not charted.
- Only unencrypted S-57 ENC cells are read; S-63 and other encrypted chart services are not. ENC survey-quality (CATZOC) information is not used.
- No tide is applied; depths are below chart datum.
- Interpolation estimates between soundings; the seabed between them is unknown.
- The look-ahead follows COG and the configured zone only.
- The alarm sounds only on the OpenCPN PC (and as $IIALR if enabled); it depends on OpenCPN running and receiving a position.
- Not type-approved navigation equipment.